

Describing Distributions & Linear Regression

ANSWER KEY



Describing shape, center, and spread

- 1 A histogram of household incomes has a long tail stretching toward high values, with most households clustered at lower incomes. Describe the shape of this distribution and state which measure of center (mean or median) is more appropriate to report, with justification.

This distribution is skewed right (the long tail points toward high values). The median is more appropriate, since the mean would be pulled upward by the small number of very high incomes, giving a misleadingly high sense of a "typical" household's income; the median is resistant to this effect.

- 2 Describe what it means for a distribution to be symmetric, and state the relationship between the mean and median in that case.

A symmetric distribution looks approximately the same on both sides of its center when folded in half. In a symmetric distribution, the mean and median are approximately equal, since there is no long tail pulling the mean away from the median.

Scatterplots and correlation

- 3 The scatterplot shows hours studied (x) versus test score (y) for 8 students, along with the least-squares regression line $\hat{y} = 52 + 6x$.

a Based on the scatterplot, describe the direction and strength of the association between hours studied and test score. The association is positive (as hours studied increases, test score tends to increase) and appears fairly strong, since the points cluster closely around the upward-sloping line.

b Use the regression equation to predict the test score for a student who studies 4.5 hours.
 $\hat{y} = 52 + 6(4.5) = 52 + 27 = 79$.

Interpreting the correlation coefficient

- 4 Explain what a correlation coefficient of $r = 0.85$ indicates about a linear relationship, and explain why r alone cannot confirm that x causes y .

$r = 0.85$ indicates a strong, positive linear relationship between the two variables — as one increases, the other tends to increase as well, and the points cluster fairly tightly around a line. However, correlation does not imply causation: a strong linear association could instead be explained by a lurking (confounding) variable influencing both, or simply by coincidence, so r alone cannot establish that changes in x directly cause changes in y .

- 5 A data set has correlation coefficient $r = -0.4$. Describe the direction and strength of the linear relationship.

The negative sign indicates a negative association (as x increases, y tends to decrease). The magnitude 0.4 indicates a weak-to-moderate linear relationship, since it is much closer to 0 than to ± 1 .

Least-squares regression and residuals

- 6 Using the regression equation $\hat{y} = 52 + 6x$ from earlier, find the residual for a student who studied 5 hours and actually scored 79.

Predicted: $\hat{y} = 52 + 6(5) = 82$. Residual = observed – predicted = $79 - 82 = -3$. The negative residual means this student scored 3 points below what the regression line predicted.

- 7 Explain what it means for a residual plot to show "no clear pattern," and why this supports using a linear model for the data.

A residual plot with no clear pattern (residuals scattered randomly above and below zero, with roughly constant spread) suggests that a linear model captures the underlying relationship well — any leftover variation is just random scatter, not a systematic trend the line failed to capture (such as a curve). If a residual plot instead showed a clear curve or fan shape, that would suggest a linear model is not appropriate.

Coefficient of determination

- 8 For the study-hours regression, $r = 0.85$. Find r^2 and interpret its meaning in context.

$r^2 = (0.85)^2 = 0.7225$, or about 72.25%. This means approximately 72.25% of the variation in test scores can be explained by the linear relationship with hours studied, leaving about 27.75% of the variation explained by other factors.

Influential points and outliers in regression

- 9 A data point has an unusually large x -value far from the rest of the data, and removing it noticeably changes the slope of the regression line. Explain why this point is called influential, and contrast it with a point that has a large residual but does not change the line much when removed.

A point is influential when its removal substantially changes the regression line (e.g. the slope or intercept) — this typically happens with points that have extreme x -values, since they have high leverage over the line's fit. A point with a large residual but ordinary x -value is simply an outlier in the y -direction; it may look unusual but doesn't have the leverage to substantially shift the line when removed, since points near the middle of the x -range have less influence on slope.

Extrapolation

- 10** Using the regression $\hat{y} = 52 + 6x$ (fit using data from students who studied between 1 and 7 hours), explain why predicting a test score for a student who studied 20 hours would be unreliable.

Predicting for $x = 20$ requires extrapolation — using the regression line far outside the range of x -values (1 to 7 hours) that was actually used to fit it. There is no guarantee the linear relationship continues to hold outside the observed data range (a student's score can't exceed 100, for instance), so such a prediction would likely be unreliable.

Multiple choice — easy 15

- 11** A least-squares regression line is $\hat{y} = 16 - 5x$. Predict $\hat{y}$ when $x = 12$.

- a. 76
- b. -46
- c. -42
- d. -44 ✓

$$\hat{y} = 16 + (-5)(12) = -44.$$

- 12** A least-squares regression line is $\hat{y} = 66 - 4x$. Predict $\hat{y}$ when $x = 17$.

- a. 0
- b. -4
- c. 134
- d. -2 ✓

$$\hat{y} = 66 + (-4)(17) = -2.$$

13 A least-squares regression line is $\hat{y} = 82 + 3x$. Predict $\hat{y}$ when $x = 1$.

- a. 87
- b. 83
- c. 79
- d. 85 ✓

$$\hat{y} = 82 + (3)(1) = 85.$$

14 A least-squares regression line is $\hat{y} = 44 - x$. Predict $\hat{y}$ when $x = 18$.

- a. 62
- b. 28
- c. 26 ✓
- d. 24

$$\hat{y} = 44 + (-1)(18) = 26.$$

15 A least-squares regression line is $\hat{y} = 11 + 8x$. Predict $\hat{y}$ when $x = 5$.

- a. -29
- b. 49
- c. 53
- d. 51 ✓

$$\hat{y} = 11 + (8)(5) = 51.$$

16 A least-squares regression line is $\hat{y} = 17 - 7x$. Predict $\hat{y}$ when $x = 12$.

- a. -69
- b. -67 ✓
- c. 101
- d. -65

$$\hat{y} = 17 + (-7)(12) = -67.$$

17 A least-squares regression line is $\hat{y} = 55 + 3x$. Predict $\hat{y}$ when $x = 13$.

a. 16

b. 96

c. 94 ✓

d. 92

$$\hat{y} = 55 + (3)(13) = 94.$$

18 A least-squares regression line is $\hat{y} = 25 - 5x$. Predict $\hat{y}$ when $x = 12$.

a. -37

b. -33

c. -35 ✓

d. 85

$$\hat{y} = 25 + (-5)(12) = -35.$$

19 A least-squares regression line is $\hat{y} = 78 - 3x$. Predict $\hat{y}$ when $x = 16$.

a. 32

b. 28

c. 30 ✓

d. 126

$$\hat{y} = 78 + (-3)(16) = 30.$$

20 A least-squares regression line is $\hat{y} = 32 - 4x$. Predict $\hat{y}$ when $x = 15$.

- a. -26
- b. -30
- c. -28 ✓
- d. 92

$$\hat{y} = 32 + (-4)(15) = -28.$$

21 A least-squares regression line is $\hat{y} = 17 - 7x$. Predict $\hat{y}$ when $x = 1$.

- a. 8
- b. 10 ✓
- c. 12
- d. 24

$$\hat{y} = 17 + (-7)(1) = 10.$$

22 A least-squares regression line is $\hat{y} = 39 - 5x$. Predict $\hat{y}$ when $x = 18$.

- a. -51 ✓
- b. -49
- c. 129
- d. -53

$$\hat{y} = 39 + (-5)(18) = -51.$$

23 A least-squares regression line is $\hat{y} = 69 + 3x$. Predict $\hat{y}$ when $x = 5$.

- a. 86
- b. 54
- c. 82
- d. 84 ✓

$$\hat{y} = 69 + (3)(5) = 84.$$

24 A least-squares regression line is $\hat{y} = 79 + 4x$. Predict $\hat{y}$ when $x = 1$.

a. 75

b. 83 ✓

c. 85

d. 81

$$\hat{y} = 79 + (4)(1) = 83.$$

25 A least-squares regression line is $\hat{y} = 62 - 8x$. Predict $\hat{y}$ when $x = 16$.

a. -64

b. -66 ✓

c. -68

d. 190

$$\hat{y} = 62 + (-8)(16) = -66.$$

Multiple choice — medium 15

26 A least-squares regression line is $\hat{y} = 52 + 2x$. For a point with $x = 2$ and observed value $y = 49$, find the residual (observed - predicted).

a. -9

b. -5

c. 7

d. -7 ✓

$$\text{Predicted: } \hat{y} = 52 + (2)(2) = 56. \text{ Residual} = y - \hat{y} = 49 - (56) = -7.$$

- 27 A least-squares regression line is $\hat{y} = 16 + 5x$. For a point with $x = 15$ and observed value $y = 84$, find the residual (observed $-$ predicted).

a. -7 ✓

b. -9

c. -5

d. 7

Predicted: $\hat{y} = 16 + (5)(15) = 91$. Residual $= y - \hat{y} = 84 - (91) = -7$.

- 28 A least-squares regression line is $\hat{y} = 56 - 9x$. For a point with $x = 14$ and observed value $y = -69$, find the residual (observed $-$ predicted).

a. 1 ✓

b. -1

c. 4

d. 3

Predicted: $\hat{y} = 56 + (-9)(14) = -70$. Residual $= y - \hat{y} = -69 - (-70) = 1$.

- 29 A least-squares regression line is $\hat{y} = 73 - 9x$. For a point with $x = 6$ and observed value $y = 29$, find the residual (observed $-$ predicted).

a. 8

b. 12

c. 10 ✓

d. -10

Predicted: $\hat{y} = 73 + (-9)(6) = 19$. Residual $= y - \hat{y} = 29 - (19) = 10$.

- 30 A least-squares regression line is $\hat{y} = 41 + 4x$. For a point with $x = 11$ and observed value $y = 82$, find the residual (observed $-$ predicted).
- a. 3
 - b. -1
 - c. -5
 - d. -3 ✓

Predicted: $\hat{y} = 41 + (4)(11) = 85$. Residual $= y - \hat{y} = 82 - (85) = -3$.

- 31 A least-squares regression line is $\hat{y} = 36 - 3x$. For a point with $x = 13$ and observed value $y = 1$, find the residual (observed $-$ predicted).
- a. 2
 - b. 4 ✓
 - c. -4
 - d. 6

Predicted: $\hat{y} = 36 + (-3)(13) = -3$. Residual $= y - \hat{y} = 1 - (-3) = 4$.

- 32 A least-squares regression line is $\hat{y} = 33 + 6x$. For a point with $x = 9$ and observed value $y = 82$, find the residual (observed $-$ predicted).
- a. -5 ✓
 - b. -7
 - c. 5
 - d. -3

Predicted: $\hat{y} = 33 + (6)(9) = 87$. Residual $= y - \hat{y} = 82 - (87) = -5$.

- 33 A least-squares regression line is $\hat{y} = 26 + 7x$. For a point with $x = 2$ and observed value $y = 41$, find the residual (observed – predicted).

a. 3

b. 1 ✓

c. 4

d. -1

Predicted: $\hat{y} = 26 + (7)(2) = 40$. Residual = $y - \hat{y} = 41 - (40) = 1$.

- 34 A least-squares regression line is $\hat{y} = 60 + 3x$. For a point with $x = 14$ and observed value $y = 111$, find the residual (observed – predicted).

a. 9 ✓

b. 11

c. -9

d. 7

Predicted: $\hat{y} = 60 + (3)(14) = 102$. Residual = $y - \hat{y} = 111 - (102) = 9$.

- 35 A least-squares regression line is $\hat{y} = 37 + 9x$. For a point with $x = 11$ and observed value $y = 128$, find the residual (observed – predicted).

a. 8

b. -8 ✓

c. -10

d. -6

Predicted: $\hat{y} = 37 + (9)(11) = 136$. Residual = $y - \hat{y} = 128 - (136) = -8$.

- 36 A least-squares regression line is $\hat{y} = 21 - 2x$. For a point with $x = 10$ and observed value $y = -4$, find the residual (observed $-$ predicted).
- a. -5 ✓
 - b. -3
 - c. 5
 - d. -7

Predicted: $\hat{y} = 21 + (-2)(10) = 1$. Residual $= y - \hat{y} = -4 - (1) = -5$.

- 37 A least-squares regression line is $\hat{y} = 26 + 4x$. For a point with $x = 6$ and observed value $y = 55$, find the residual (observed $-$ predicted).
- a. 7
 - b. 5 ✓
 - c. 3
 - d. -5

Predicted: $\hat{y} = 26 + (4)(6) = 50$. Residual $= y - \hat{y} = 55 - (50) = 5$.

- 38 A least-squares regression line is $\hat{y} = 64 + 7x$. For a point with $x = 5$ and observed value $y = 101$, find the residual (observed $-$ predicted).
- a. 4
 - b. 0
 - c. 2 ✓
 - d. -2

Predicted: $\hat{y} = 64 + (7)(5) = 99$. Residual $= y - \hat{y} = 101 - (99) = 2$.

- 39 A least-squares regression line is $\hat{y} = 79 + x$. For a point with $x = 3$ and observed value $y = 70$, find the residual (observed $-$ predicted).
- a. -14
 - b. -10
 - c. -12 ✓
 - d. 12

Predicted: $\hat{y} = 79 + (1)(3) = 82$. Residual $= y - \hat{y} = 70 - (82) = -12$.

- 40 A least-squares regression line is $\hat{y} = 45 - 9x$. For a point with $x = 15$ and observed value $y = -91$, find the residual (observed $-$ predicted).
- a. -1 ✓
 - b. 1
 - c. -4
 - d. -3

Predicted: $\hat{y} = 45 + (-9)(15) = -90$. Residual $= y - \hat{y} = -91 - (-90) = -1$.

Multiple choice — hard 15

- 41 A regression has correlation coefficient $r = -0.85$. Find r^2 , expressed as a percentage, to two decimal places.
- a. 74.25%
 - b. 85.00%
 - c. 70.25%
 - d. 72.25% ✓

$r^2 = (-0.85)^2 = 0.7225$, or 72.25%.

- 42 A regression has correlation coefficient $r = -0.65$. Find r^2 , expressed as a percentage, to two decimal places.

a. 42.25% ✓

b. 44.25%

c. 40.25%

d. 65.00%

$$r^2 = (-0.65)^2 = 0.4225, \text{ or } 42.25\%.$$

- 43 A regression has correlation coefficient $r = -0.80$. Find r^2 , expressed as a percentage, to two decimal places.

a. 66.00%

b. 80.00%

c. 62.00%

d. 64.00% ✓

$$r^2 = (-0.80)^2 = 0.6400, \text{ or } 64.00\%.$$

- 44 A regression has correlation coefficient $r = -0.20$. Find r^2 , expressed as a percentage, to two decimal places.

a. 20.00%

b. 4.00% ✓

c. 6.00%

d. 2.00%

$$r^2 = (-0.20)^2 = 0.0400, \text{ or } 4.00\%.$$

- 45 A regression has correlation coefficient $r = 0.50$. Find r^2 , expressed as a percentage, to two decimal places.
- a. 50.00%
 - b. 25.00% ✓
 - c. 27.00%
 - d. 23.00%

$$r^2 = (0.50)^2 = 0.2500, \text{ or } 25.00\%.$$

- 46 A distribution has mean 34 and median 47. Based on this comparison, classify the shape of the distribution.
- a. Skewed right (mean noticeably greater than median)
 - b. Cannot be determined from the mean and median alone
 - c. **Skewed left (mean noticeably less than median) ✓**
 - d. Approximately symmetric (mean and median are close)

Mean = 34, median = 47. Skewed left (mean noticeably less than median).

- 47 A distribution has mean 94 and median 82. Based on this comparison, classify the shape of the distribution.
- a. Approximately symmetric (mean and median are close)
 - b. **Skewed right (mean noticeably greater than median) ✓**
 - c. Skewed left (mean noticeably less than median)
 - d. Cannot be determined from the mean and median alone

Mean = 94, median = 82. Skewed right (mean noticeably greater than median).

48 A distribution has mean 71 and median 63. Based on this comparison, classify the shape of the distribution.

- a. Skewed left (mean noticeably less than median)
- b. Cannot be determined from the mean and median alone
- c. Approximately symmetric (mean and median are close)
- d. Skewed right (mean noticeably greater than median) ✓**

Mean = 71, median = 63. Skewed right (mean noticeably greater than median).

49 A distribution has mean 83 and median 84. Based on this comparison, classify the shape of the distribution.

- a. Approximately symmetric (mean and median are close) ✓**
- b. Skewed right (mean noticeably greater than median)
- c. Skewed left (mean noticeably less than median)
- d. Cannot be determined from the mean and median alone

Mean = 83, median = 84. Approximately symmetric (mean and median are close).

50 A distribution has mean 46 and median 40. Based on this comparison, classify the shape of the distribution.

- a. Skewed left (mean noticeably less than median)
- b. Cannot be determined from the mean and median alone
- c. Skewed right (mean noticeably greater than median) ✓**
- d. Approximately symmetric (mean and median are close)

Mean = 46, median = 40. Skewed right (mean noticeably greater than median).

- 51 A regression has coefficient of determination $r^2 = 0.49$. Find the possible value(s) of the correlation coefficient r .

- a. ± 0.80
- b. ± 0.70 ✓
- c. 0.70
- d. ± 0.49

$r = \pm\sqrt{0.49} = \pm 0.70$ (since r^2 does not determine the sign of r , both the positive and negative square root are possible).

- 52 A regression has coefficient of determination $r^2 = 0.64$. Find the possible value(s) of the correlation coefficient r .

- a. ± 0.64
- b. 0.80
- c. ± 0.90
- d. ± 0.80 ✓

$r = \pm\sqrt{0.64} = \pm 0.80$ (since r^2 does not determine the sign of r , both the positive and negative square root are possible).

- 53 A regression has coefficient of determination $r^2 = 0.36$. Find the possible value(s) of the correlation coefficient r .

- a. ± 0.60 ✓
- b. 0.60
- c. ± 0.36
- d. ± 0.70

$r = \pm\sqrt{0.36} = \pm 0.60$ (since r^2 does not determine the sign of r , both the positive and negative square root are possible).

- 54 A regression has coefficient of determination $r^2 = 0.16$. Find the possible value(s) of the correlation coefficient r .

- a. ± 0.16
- b. ± 0.50
- c. 0.40
- d. ± 0.40 ✓

$r = \pm\sqrt{0.16} = \pm 0.40$ (since r^2 does not determine the sign of r , both the positive and negative square root are possible).

- 55 A regression has coefficient of determination $r^2 = 0.04$. Find the possible value(s) of the correlation coefficient r .

- a. ± 0.20 ✓
- b. ± 0.04
- c. 0.20
- d. ± 0.30

$r = \pm\sqrt{0.04} = \pm 0.20$ (since r^2 does not determine the sign of r , both the positive and negative square root are possible).